

CSCE 190: Computing in the Modern World

Class 5 – GitHub Pages – Example of *Resources* (HTML/CSS, Markdown) via GitHub

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SEP 2026

Carolinian Creed: “I will practice personal and academic integrity.”

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Outline

- Introduction Section
 - Recap from Last Class
- Main Section
 - Web Sites, Web Pages
 - GitHub Pages – example of sharing, HTML/CSS and Markdown
- Conclusion Section
 - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 4 + Lab

We talked about

- Sharing Resources – Content Format
- HTML, CSS
- Markdown
- GitHub pages

Activity 4 – HTML / CSS, Markdown

- Objective A: learn HTML/ CSS
- Tasks
 - See HTML cheat sheet
 - Create HTML files with an editor (like Notepad, vi) or online tool
 - Learn about different constructs
- Objective B: learn Markdown
- Tasks
 - See Markdown cheat sheet
 - Create Markdown files with an editor (like Notepad, vi)
 - Learn about different constructs

Recent Discussion – AI and Safety

“Researchers have long warned that three conditions could lead to loss of control:

- a misaligned goal,
- the capability to pursue it, and
- an environment that allows it,”

ACM A.M. Turing Award laureate Yoshua Bengio said at UN
(Sep 2026)

<https://thehill.com/policy/technology/6102955-un-ai-panel-urges-safeguards/>



Main Section

Concept: Web Site, Web Page

- An online site, identified by a URL, consisting of one or more web pages
- Pre-requisites for a web site
 - A web address: *commonly* called getting a domain address
 - Site content: web pages
- Web page can be created by a website editor
 - GitHub pages
 - Google sites
 - Online tools – Wix, GoDaddy, ...
 - Editors – **Both WYSIWYG and Typesetting choices**

Editor Concept: Format and Content

WYSIWYG – what you see is what you get

- Software that lets you edit content so it looks just like the final result (format)
- Pros
 - Easy to edit
- Cons
 - If format changes slightly, may spend a lot of time fixing it, even if the content is same

Examples

- Word Processors: Programs like Microsoft Word or Google Docs.
- **Website Builders: Tools like Wix or WordPress where you drag and drop items.**
- Email Clients: Gmail or Outlook when you change text color or add links

Editor Concept: Format and Content

Typesetting – separate content from format

- Software that lets you edit content separately;
- Format can be changed separately as well
- Pros
 - Easy to convert format
 - Advanced styles can be used seamlessly
- Cons
 - Harder to use software

Examples

- Adobe InDesign: The professional industry standard for books, magazines, and complex layouts.
- LaTeX: A code-based system ideal for heavy academic, mathematical, and technical documents.
- **HTML/ CSS: html for content, css for digital styling**

Default file: index.html

- At a web address, index.html (or index.htm) is the file that a browser looks by default
- Otherwise, the name of a file has to be explicitly mentioned as part of a URL call
 - Example: <https://biplav-s.github.io/> (file not mentioned; index.html is assumed)
- Question: what happens if file is mentioned ?
 - The named file is opened
 - Example: <https://github.com/biplav-s/biplav-s.github.io/blob/master/index.html>

Default file: Readme.md

- On GitHub, the file that is looked by default is Readme.md
- Example: <https://github.com/biplav-s/biplav-s.github.io/>
 - File opened: Readme.md

Creating One's GitHub Page

- Have a GitHub account, say ExampleAccount
- A website for you is automatically provided: ExampleAccount.github.io
- Default location for website content is provided:
 - ExampleAccount.github.io/ExampleAccount/ExampleAccount.github.io
 - Create a sample index.html

Sample index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>CSCE 190: [your name]</title>
</head>

<body>
  <h1> [your name] </h1>
  <h3> [your name] – Hello World!</h3>
</body>
</html>
```

Content of index.html

Adapted from: Robert Bailey's presentation

Example 2: TA 1's Website

A simple Fun Fact website using Github Pages

Link: https://apallavias04-source.github.io/DemoPage_class190/

apallavias04-source.github.io/DemoPage_class190/



HTML elements used

<!DOCTYPE html> — Declaration (not a tag) that tells the browser to render the page using modern HTML5 rules.

<html> — The root element that wraps the entire page.

<head> — Holds metadata about the page that isn't shown directly on screen.

<meta> — Supplies metadata like character encoding and viewport settings; here it sets UTF-8 encoding and makes the page mobile-responsive.

<title> — Sets the text shown in the browser tab; here it's "Welcome."

<style> — Contains the page's CSS; here it defines the background image, card styling, and button look.

<body> — Holds everything visible on the page.

<div> — A generic container with no inherent meaning; used twice here - once for the glass .card wrapper, once for the clickable #box.

<h1> — The top-level heading; displays "Welcome."

<script> — Contains the page's JavaScript; here it holds the facts array and the **showFact()** function that swaps in a random fact on click.

Example 3: TA 2's Website

A personal academic portfolio website hosted on GitHub Pages with a custom domain

Link: <https://aataiwo.com>

Source Code: <https://github.com/Akian-coder/portfolio>



HTML elements used

`<!DOCTYPE html>` — Tells the browser to use modern HTML5.

`<html>` — The main container for the entire website.

`<head>` — Contains page information, fonts, styling, and the website title.

`<meta>` — Sets character encoding, mobile responsiveness, and site description.

`<style>` — Contains the CSS used for the dark theme, navigation bar, responsive layout, cards, and buttons.

`<body>` — Contains all visible website content.

`<header>` and `<nav>` — Create the top navigation menu for Home, About, Skills, Research, Projects, and Contact.

`<section>` — Organizes the website into different content areas.

`<a>` — Creates links to sections of the website and external platforms such as GitHub and LinkedIn.

`<img>` — Displays images, including the profile image and project images.

Discussion – Audience, Content, Elements

- Audience: decide who will see the website / webpage
- Content: decide what content to put
- Page Elements:
 - Simple is better. Have a reason to include a feature
 - Keep it mobile friendly
 - Beyond content, focus on web page loading performance
 - Keep evolving over time

Assignment 5

- Assignment
- Create a Github account
- Create a github.io repository on Github
- Connect to your repo and Commit a change
- Using VS Code, the Github website, Github dev, etc.
- Test your github.io link

Concluding Section

Conclusion

We talked about

- Web Sites, Web Pages
- GitHub Pages – example of sharing, HTML/CSS and Markdown

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics) + Lab	Github account, repo
4	HTML basics, IDE (VS Code)	None
5	More HTML, CSS, Problem Solving	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	AI – LLMs/FMs, Generative AI, Issues and Challenges	Problem Solving with GenAI
10	Building a resume, LinkedIn	Draft resume in PDF
11	Review resumes	Review resumes, update Github
12	Convert resume to HTML	Begin conversion, update Github
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	All extensions end

Ask Me Anything
